

1. **Design a GUI application that allows the user to maintain a simple student record (add /update/delete/display) using appropriate layout manager, controls, and event handling.**
Code:

```
import javax.swing.*;
import javax.swing.table.DefaultTableModel;
import java.awt.*;

public class Q10_StudentRecordSystem extends JFrame {
    JTextField nameField = new JTextField();
    JTextField rollField = new JTextField();
    JTextField courseField = new JTextField();
    DefaultTableModel model = new DefaultTableModel(
        new String[]{"Name", "Roll No", "Course"}, 0);
    JTable table = new JTable(model);

    Q10_StudentRecordSystem() {
        setTitle("Student Record Management System");
        setSize(750, 500);
        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        setLocationRelativeTo(null);

        JPanel form = new JPanel(new GridLayout(2, 6, 10, 10));
        JButton add = new JButton("Add");
        JButton update = new JButton("Update");
        JButton delete = new JButton("Delete");

        form.add(new JLabel("Name")); form.add(nameField);
        form.add(new JLabel("Roll No")); form.add(rollField);
        form.add(new JLabel("Course")); form.add(courseField);
        form.add(add); form.add(update); form.add(delete);

        add(form, BorderLayout.NORTH);
        add(new JScrollPane(table), BorderLayout.CENTER);

        add.addActionListener(e -> addStudent());
        update.addActionListener(e -> updateStudent());
        delete.addActionListener(e -> deleteStudent());

        table.getSelectionModel().addListSelectionListener(e -> {
            int row = table.getSelectedRow();
            if (row != -1) {
                nameField.setText(model.getValueAt(row, 0).toString());
                rollField.setText(model.getValueAt(row, 1).toString());
                courseField.setText(model.getValueAt(row, 2).toString());
            }
        });
    }

    void addStudent() {
        if (nameField.getText().isEmpty() || rollField.getText().isEmpty()
            || courseField.getText().isEmpty()) {
            JOptionPane.showMessageDialog(this, "Enter all details");
            return;
        }
    }
}
```

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    }

    model.addRow(new Object[]{
        nameField.getText(), rollField.getText(), courseField.getText()
    });
    clearFields();
}

void updateStudent() {
    int row = table.getSelectedRow();
    if (row == -1) {
        JOptionPane.showMessageDialog(this, "Select a student");
        return;
    }

    model.setValueAt(nameField.getText(), row, 0);
    model.setValueAt(rollField.getText(), row, 1);
    model.setValueAt(courseField.getText(), row, 2);
    JOptionPane.showMessageDialog(this, "Student Updated");
}

void deleteStudent() {
    int row = table.getSelectedRow();
    if (row == -1) {
        JOptionPane.showMessageDialog(this, "Select a student");
        return;
    }
    model.removeRow(row);
    clearFields();
}

void clearFields() {
    nameField.setText("");
    rollField.setText("");
    courseField.setText("");
}

public static void main(String[] args) {
    SwingUtilities.invokeLater(() -> new Q10_StudentRecordSystem().setVisible(true));
}
}

```

Output:

